

A Mechanistic, Numerical Model of Sedimentation, Mineralization, and Decomposition for Marsh Sediments¹

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ABSTRACT

This paper describes a computer model that relates measured nutrient profiles to organic matter diagenesis in marsh sediments. Processes included in the model are sedimentation of exogenous and endogenous organic and inorganic matter, decomposition, above- and belowground biomass and production, and N and P mineralization. The shape of organic nutrient profiles in marsh sediments is controlled by the relative rates of nutrient mineralization and inputs of organic and inorganic matter. Sediment inputs were specified for a freshwater, tidal marsh on the North River, Massachusetts, and the model was solved for nutrient mineralization rates by fitting the model to observed nutrient profiles. Model calculations of net annual N and P export from a 1-m sediment column, 9.6 to 17.5 g N m⁻² yr⁻¹ and 1.19 to 2.02 g P m⁻² yr⁻¹, were sensitive to small changes in estimates of belowground production and the fraction of organic material that is refractory, constants in the model that are poorly known for marsh ecosystems. These export rates are about 55 to 78% and 71 to 85% of the annual sediment inputs of organic N and P, respectively. The fate of nutrients exported from the sediment was not specified, but it could include plant uptake, denitrification, and net loss to the adjacent tidal river. The model demonstrates that mature marshes with deep sediments recycle proportionally more of the nutrients required by primary producers than young marshes

with shallow sediments. Model calculations indicated that the greatest changes in C, N, and P occur in the upper 2 cm of sediment. Labile organic C decomposed most rapidly which caused the relative concentrations of organic N and P to increase. Below this thin surface zone C loss became negligibly small, but there was a gradual decline in concentrations of organic N and P as they continued to mineralize slowly within the upper 60 cm of sediment. Apparently, only refractory C, N, and P compounds remained below the 60-cm depth. The predicted ratio of N/P loss from sediment ranged from 8.2 N/1 P to 9.55 N/1 P (g/g) and was constant over depth; this compares to a ratio of about 11 N/1 P in litter of the dominant macrophytes. This model is sufficiently general as to be useful in studies of sediment formation and nutrient cycling in other systems.

Additional Index Words: N, P, C, diagenesis, sediments, nutrient cycling, wetlands.

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THE VERTICAL DISTRIBUTION of nutrients in the sediment of a typical undisturbed marsh is a product of several processes that proceed on different time scales: plant production, sedimentation, and microbial mineralization of organic matter. For example, it may require well over 100 yr to accumulate 1 m of sediment from the litter of marsh vegetation (Arman-tano and Woodwell, 1975; Harrison and Bloom, 1974; DeLaune et al., 1978). In comparison, decomposition

of marsh plant material may be functionally complete in weeks to months (Odum and Heywood, 1978; Marinucci, 1982; Boyd, 1970; Montagna and Ruber, 1980) while nutrient regeneration from plant litter can be measured in hours (Bowden, 1984a; Henricksen and Jensen, 1979). Although these processes have been measured in a variety of ecosystems, we are unaware of any study that provides estimates for all three processes and relates these to sediment formation and nutrient cycling.

The purpose of this paper is to describe a numerical model (SEMIDEC) that integrates the effects of sedimentation, nutrient mineralization, and decomposition of organic matter in a marsh sediment. The unique feature of this model is its treatment of sediment horizons as year-class cohorts which are modified over time as they are buried. Assuming that inputs are in steady state, time and depth dependent changes in a cohort are used to simulate depth profiles of total sediment C, N, and P.

This approach differs from the models developed by Berner (1971; 1980), who described the process of sediment diagenesis with a series of differential equations that calculate instantaneous fluxes of nutrients from sediments based on concentration gradients. Our model does not use diffusion explicitly; instead, it computes annual nutrient losses based on the concentrations of organic nutrients that remain in subsurface soils. Furthermore, Berner's models are restricted to sediments where pore water advection and plant root activity are negligible. These processes are significant in the marsh environment and must be considered, as our model does.

Our model also differs from that of Hsieh and Webster (1984), the only other model we are aware of that describes sediment accumulation rates in tidal marshes. Their model describes soil organic matter accumulation, but it ignores mineral accumulation which may be the major component of sedimentation in some marshes (DeLaune et al., 1978). Furthermore, the Hsieh and Webster model allows only the accumulation of aboveground primary production and ignores the large contribution due to belowground production. Our model accounts for mineral accumulation and belowground production.

Although the data used to develop and test our model were acquired for a tidal fresh-to-brackish water marsh in Massachusetts, the model has wide applicability. The processes included in the model are sufficiently general that, with appropriate modifications to the model parameters, it could be used to describe sediment profiles and to calculate nutrient turnover in other marsh or soil systems.

Research Site

The research site was a 24-ha tidal, freshwater marsh located on the North River about 20-km upstream from the river's mouth on Cape Cod Bay. The marsh sediment consists of up to 1 m of peat on glacial till. The original soil survey (Upham, 1969) classified this soil only as a peat. However, P. Fletcher (personal communication, 1985) classifies it as a Dysic, mesic Typic Medisapristis Freetown muck. Sediment water content (88% of fresh weight) and organic content (60% of dry weight) were relatively constant from

the surface to at least the 70-cm depth. The sediments were saturated as a result of tidal flooding which occurred twice daily during seasons of high freshwater discharge or sporadically during dry seasons. The marsh currently supports a diverse plant community of over 60 species. The dominant plants on a biomass basis are *Typha latifolia*, *Carex lacustris*, and *Calamagrostis canadensis*. Maximum standing biomass was about 1-kg dry wt m⁻². The majority of the aboveground plant production appeared to decompose in place with little or no export of litter to the adjacent river.

MATERIALS AND METHODS

Sediment Sampling

Sediment cores were obtained with 6.1 cm or 9.6 cm i.d. by 1-m polyvinyl chloride (PVC) core tubes that were fitted with pistons to minimize sediment compaction. In total, 20 sediment cores were removed, sectioned in the field into 2- to 5-cm lengths, and stored on ice until analysis. Each section was dried at 70°C and weighed to get bulk density. The sections were then ground with a Wiley Mill, and a portion of the sediment from each section was combusted at 500°C overnight to obtain ash content and ash-free dry weight (AFDW) of the sediment. Other portions were used to determine N and P contents, as described below.

Plant Litter Decomposition Rates

Decomposition rates of plant litter were determined using litter bags. Standing dead leaves from each of the three dominant plants (*Typha*, *Carex*, and *Calamagrostis*) were collected in November 1979 just after the leaves had senesced. The litter was sorted by species and ca 20-g portions of the sorted material were placed in 2-mm mesh fiberglass bags. The bags were deployed on the marsh surface in January 1980 at two locations about 10-m inland from the stream bank in typical stands of *Typha*, *Carex*, and *Calamagrostis*. Between five and 10 bags containing litter of each species were sampled monthly for 2.5 yr. After each sampling, the contents of each bag were washed to remove accumulated debris, dried at 70°C, and weighed. The samples were ground with a Wiley Mill, and subsamples were used to determine ash-free dry weight and total N and P contents.

Nutrient Analyses

Samples of sediments and plant tissues were digested in a block digester (Technicon BD-40) and analyzed for total phosphorus (as PO₄) and total Kjeldahl N (as NH₄⁺) by autoanalyzer (Technicon, 1977). Concentrations of NH₄⁺ and PO₄ prior to digestion were negligible compared to organic N and P so these analyses estimate total organic N (TON) and total organic P (TOP). For each determination, approximately 2 g of dry sediment or dry litter (70°C) was digested for 2 h at 400°C with 25 mL of H₂SO₄ and a catalyst (Kel-Pak, Formula 1, CMS), then diluted to 250 mL before analysis.

MODEL DESCRIPTION

The model uses plant litter-fall, root turnover, allochthonous sedimentation of organic and inorganic matter, degradation of bulk organic matter, and mineralization of TON and TOP to simulate marsh sediment formation. The key feature of this model is that it is a single-year cohort model; that is, it simulates diagenesis of a single year class of sediment under a unit cm² of marsh surface. Christian (1984) used a similar life-history approach to describe the decomposition of litter.

The model computes depth profiles of sediment organic matter and concentrations of TON and TOP from the integrated behavior of a single cohort over time, and it predicts

Table 1. Definitions and best estimates of parameters.

Symbol	Definition	Value
k	Specific decay constant for organic matter†	0.7142 yr ⁻¹
k_1	A distribution parameter for belowground biomass‡	0.1 cm ⁻¹
k_2	Turnover rate of live belowground biomass§	0.5 yr ⁻¹
k_3	Concentration of ash in plant matter†	0.0344 g g ⁻¹
k_4	Annual allochthonous input of inorganic matter¶	0.02 g cm ⁻²
k_5	Nitrogen concentration in leaf litter†	0.0074 g g ⁻¹
k_6	Specific loss rate of soluble N¶	12.0 yr ⁻¹
k_7	Nitrogen concentration in live belowground biomass†	0.02 g g ⁻¹
k_8	Particle density of organic matter#	1.44 g cm ⁻³
k_9	Particle density of inorganic matter#	2.5 g cm ⁻³
k_{10}	Phosphorus concentration in live belowground biomass†	0.002 g g ⁻¹
k_{11}	Ratio of losses of N to P¶	8.2 g g ⁻¹
b_o	Bulk density of organic matter†	0.07182 g cm ⁻³
B	Net annual dry aboveground plant production†	0.1 g cm ⁻²
f_c	Fraction of organic matter that is refractory†	0.2 g g ⁻¹
f_n	Fraction of organic nitrogen that is refractory¶	0.45 g g ⁻¹
m	Specific mineralization rate of labile organic N¶	0.6 yr ⁻¹
$P_o(0)$	Annual deposition of organic P onto marsh surface†	6.73×10^{-5} g cm ⁻²
r	Residence time of soluble N in cohort¶	0.0025 yr
R_o	Live belowground biomass at sediment surface‡	0.0105 g cm ⁻²
R_l	Integrated live belowground biomass‡	0.1 g cm ⁻²
$W_a(0)$	Annual deposition of organic matter onto marsh surface†	0.1 g cm ⁻²

Date sources:

† Independent measurements.

‡ Biomass data obtained from cultures, the depth distribution is assumed.

§ Assumption.

¶ Gives a best fit of the model.

CRC Handbook (1978), densities of cellulose and quartz.

the annual turnover of N and P from a sediment column of a given age or depth by integrating over time the annual losses from the cohort. These simulated profiles are valid if the inputs to the sediment surface remain constant during the time course of the simulation. A discontinuity in the vertical distribution of a given variable would indicate that the steady-state assumption is invalid. However, the steady-state assumption can be relaxed to explore the consequences of different management alternatives or to include specific events (significant storms, etc.) if the history of a site is known. The model can not be used to simulate nutrient dynamics in a sediment where bioturbation alters the sediment profile or in soils that have been tilled.

Definitions for the variables and parameters used in the model and in the following discussion are listed in Tables 1 and 2.

Organic Inputs and Decomposition

There are three primary inputs of organic matter to a sediment cohort. Surface inputs may come from plant litter produced in situ or from riverine detritus. The model also

Table 2. Definitions and dimensions of variables.

Symbol	Definition	Dimension
$C_l(t)$	Amount of labile organic matter in the cohort	g dry wt
$C_r(t)$	Amount of refractory organic matter in the cohort	g dry wt
$C_T(t)$	Total amount of organic matter in the cohort	g dry wt
D_u	Depth from sediment surface to top of cohort	cm
D_b	Depth from sediment surface to bottom of cohort	cm
$G(t)$	Gross mineralization rate of organic N within cohort	g N cm ⁻² yr ⁻¹
I_n	Total annual input of organic N into sediment	g N cm ⁻²
M_n	Total annual net export of N from the sediment between the surface and depth D_b	g N cm ⁻²
$N_i(t)$	Concentration of soluble N in cohort	g N mL H ₂ O ⁻¹
$N_l(t)$	Total amount of labile organic N in cohort	g cm ⁻²
$N_r(t)$	Total amount of refractory organic N in cohort	g cm ⁻²
$N_T(t)$	Total amount of organic N in cohort	g cm ⁻²
$P_o(t)$	Total amount of organic P in cohort	g cm ⁻²
$V(t)$	Wet cohort volume	cm ³
$W_i(t)$	Amount of inorganic matter in cohort	g cm ⁻²
$W_b(t)$	Amount of live biomass in cohort	g cm ⁻²

allows inputs of organic matter within the sediment as a result of the turnover and decomposition of belowground plant parts. The model adds these within-sediment inputs to the sediment cohort as functions of sediment depth and root turnover rate.

We have assumed that there are two pools of organic matter: a refractory one with negligible decay and a labile pool with relatively rapid decay. This is consistent with results obtained from a litter bag study of the three dominant macrophytes in the North River Marsh (see Results). Differences in the decomposition rates of labile and refractory pools of organic matter have been recognized previously (Minderman, 1968; Howard and Howard, 1974; Lousier and Parkinson, 1976; Bunnell et al., 1977; Hunt, 1977; Jenkinson, 1977; Godshalk and Wetzel, 1978; Martin et al., 1980; Heal et al., 1981; Carpenter, 1982; Wieder and Lang, 1982; Molina et al., 1983). We defined the amount of labile organic matter added to the cohort initially in litterfall as

$$C_l(0) = (1 - f_c)W_a(0) \quad [1]$$

where f_c is the refractory fraction of the organic matter input. This refractory organic matter can be thought of as the fraction that is incorporated into humus and, since the residence time of humus is on the order of millennia (Wada and Aomine, 1973; Jenkinson and Rayner, 1977), we have assumed that on the time scale of our simulations that this fraction does not decompose. $W_a(0)$ is the annual deposition of organic matter onto the marsh surface (g cm⁻²), and we have assumed in simulations reported here that W_a is constant. However, $W_a(0)$ could be made a function of time for other applications. Note that as time increases, the age of the cohort and its depth in the sediment increase (Fig. 1).

The distribution of belowground plant parts is required to account for belowground litter production and was assumed to follow an exponential distribution over depth (Fig. 1) with the greatest concentration of biomass at the sediment surface

$$R = R_o \exp(-k_1 D) \quad [2]$$

where R_o (g cm⁻²) is the weight of roots at the sediment

surface, k_1 is a distribution parameter, and D is sediment depth (cm). Such a distribution agrees well with the observations of Valiela et al. (1976). Alternative equations could be used to simulate different root profiles.

The integrated root biomass (R_I) over all depths is defined by the integral of Eq. [2]

$$R_I = R_o \int \exp(-k_1 D) dD. \quad [3]$$

To obtain the parameters R_o and k_1 we specified a total belowground biomass, assumed that all live belowground biomass was contained within the top 30 cm of sediment, then solved Eq. [3] for R_o and k_1 .

Given these values, the weight of live belowground plant parts (W_b) within the cohort was calculated as

$$W_b(t) = R_o \int_{D_u}^{D_b} \exp(-k_1 D) dD \\ = R_o [\exp(-k_1 D_b) - \exp(-k_1 D_u)] / -k_1 \quad [4]$$

where D_b and D_u are the depths (cm) to the bottom and top of the cohort, respectively. At time zero, $D_u = 0$, and D_b is calculated from Eq. [13]. After each time iteration of 1 yr, D_u is given the value of D_b and a new D_b is calculated. By definition, D_u and D_b are separated in time by exactly 1 yr.

Given the inputs of labile aboveground litter (Eq. [1]) and the amount of belowground biomass (Eq. [4]), the total dry weight of labile organic matter in the cohort (C) is

$$C(t+1) = C(t) - kC(t) + k_2(1 - f_c)W_b(t) \quad [5]$$

where k is the decomposition rate constant, k_2 is the turnover time of belowground plant parts (yr^{-1}), and all other variables are as defined earlier. We have assumed that the fraction of organic material that is refractory is the same for both above and belowground biomass.

The dry weight of refractory organic matter in the cohort (C_r) increases as refractory belowground organic matter is added

$$C_r(t+1) = C_r(t) + k_2 f_c W_b(t) \quad [6]$$

where

$$C_r(0) = f_c W_a \quad 7$$

The total dry weight of organic matter in the cohort (C_T) is equal to the sum of labile and refractory components and live root weight:

$$C_T(t) = C(t) + C_r(t) + W_b(t). \quad [8]$$

Inorganic Inputs and Dynamics

The amount of inorganic material added to the cohort initially in plant litter and in exogenous sediment is

$$W_i(0) = k_3 W_a(0) + k_4 \quad [9]$$

where k_3 is the ash content of plant biomass (g g^{-1} , dry weight basis) and k_4 is the annual deposition of inorganic material (g cm^{-2}) from exogenous sources.

The quantity of inorganic matter within the cohort also changes with time in response to belowground plant dynamics. We assumed that inorganic material (as ash from plant matter) is removed from the cohort by root uptake to support plant production. The amount of inorganic material removed to support total aboveground production is apportioned to each cohort by the relative amount of the total root biomass in that cohort. Note that root uptake and root death are equivalent in a given cohort so that only aboveground production needs to be apportioned

$$W_i(t+1) = W_i(t) + k_3 W_b(k_2 - B/R_I) \quad [10]$$

where B is the annual net aboveground plant production (g cm^{-2}). Field observations at our study site indicate that that

$B = W_a(0)$; that is, import or export of primary production is negligible. For any application of the model, net primary production and the import or export of primary production must be determined in order to assign values to B and $W_a(0)$.

Sedimentation, Cohort Volume, and Depth

The volume of a marsh sediment with high organic matter content is determined entirely by the bulk density of the organic matter; mineral material adds mass but not volume (Gosselink et al., 1984). Thus, the volume (V) of a cohort of sediment can be computed as

$$V(t) = [C(t) + C_r(t)]/b_o + W_b(t)/b_o \quad [11]$$

where b_o is the bulk density of organic material within the cohort. For mineral soils, this equation must be modified to account for the volume of the inorganic material by including the term $W_i(t)/b_i$, where b_i is the bulk density of the inorganic soil matter.

By definition each cohort represents 1 yr accumulation of organic and inorganic matter and has a cross sectional area of 1 cm^2 . Thus, the volume of a cohort is also its vertical height, and the depth to the bottom of the cohort (D_b) can be calculated as:

$$D_b = D_u + V(t)(1 \text{ cm cm}^{-3}). \quad [12]$$

However, the biomass of belowground plant parts (W_b) and the cohort volume are interdependent; a change in one necessitates a change in the other (Fig. 1). To solve for cohort depth (and volume) we substituted equations [4] and [11] into [12] to give

$$D_b(i+1) = D_u + [C_i(t+1) + C_r(t+1)]/b_o \\ + R_o [\exp(-k_1 D_b(i)) - \exp(-k_1 D_u)] / (-k_1 b_o) \quad [13]$$

which was solved numerically for a new value of D_b corresponding to the new cohort depth. The iteration step, where (i) refers to the iteration, was repeated until the absolute difference between iterations $|D_b(i+1) - D_b(i)|$ was < 0.001 . Then this new depth was used to calculate the biomass of belowground plant parts (Eq. [4]), the volume of the cohort (Eq. [11]), and equations that contain W_b and V . We have assumed that there is no compaction of sediment with depth of burial up to 1 m. This is consistent with our observations of bulk density in North River sediments as well as those of Gosselink et al. (1984) of marsh soils in Louisiana.

Nitrogen Dynamics

Nitrogen cycles within the cohort between pools of labile organic (N_l) and soluble nitrogen (N_s), and is lost from the cohort from the soluble pool (Fig. 1). We did not distinguish between soluble N lost as inorganic or dissolved organic N. Only the net rate of loss of total N, from such processes as absorption by plants, denitrification, volatilization, or advection and diffusion from sediments, can be inferred from model calculations of "soluble N loss".

Following the arguments used for total organic matter, the annual deposition of labile organic N onto the marsh surface [$N_l(0)$] was assumed to be a constant fraction ($1 - f_n$) of the input of total organic nitrogen [$N_T(0)$] contained in leaf litter

$$N_l(0) = (1 - f_n)k_3 W_a(0) \quad [14]$$

where k_3 is the concentration (g N g dry wt^{-1}) of organic N in leaf litter. The remaining fraction of the organic N input [$N_r(0)$] was assumed to be refractory

$$N_r(0) = f_n k_3 W_a(0). \quad [15]$$

The amounts of labile and refractory organic N change over time in response to mineralization and subsequent loss

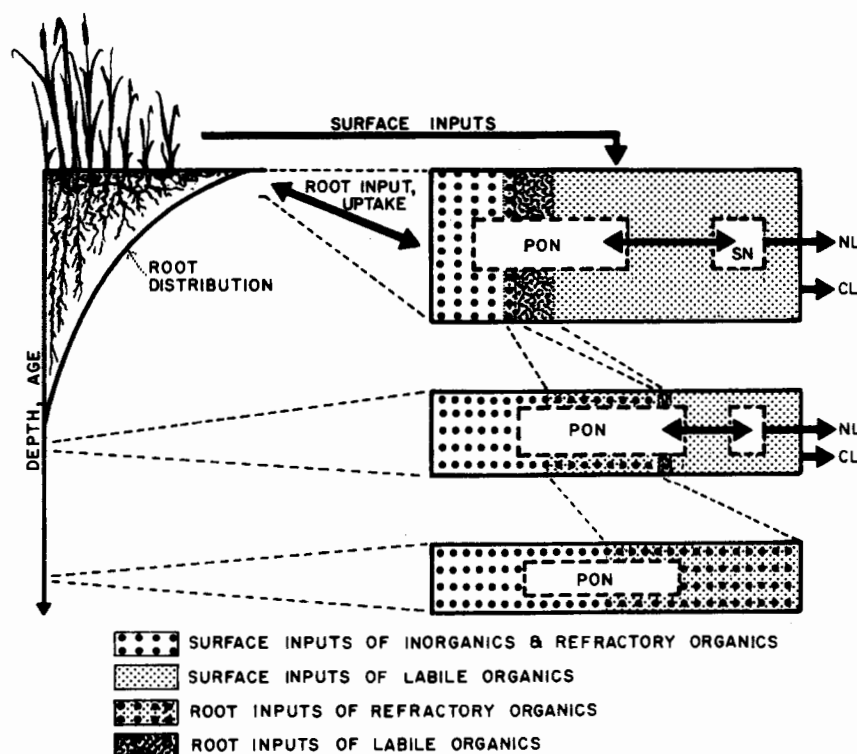


Fig. 1. Conceptualization of the dynamics of sediment cohorts showing changes in the pool sizes of labile and refractory materials as a function of depth-PON: particulate organic N; SN: soluble N; NL: net N export from sediment cohort; CL: loss of labile C compounds from sediment cohort.

of soluble N, and to litter inputs from dying roots and rhizomes.

As in other studies (Stanford and Smith, 1972; Smith et al. 1980), we assumed that N mineralization follows first order kinetics such that the rate is proportional to the amount of labile organic N

$$G(t) = mN(t) \quad [16]$$

where $G(t)$ is the gross mineralization rate within the cohort (g N yr^{-1}) and m is the specific mineralization rate (yr^{-1}). Nitrogen mineralization is temperature dependent (Stanford et al., 1973) and, thus, changes seasonally. However, we are not concerned with seasonal dynamics since the model is used to estimate annual nutrient mineralization and turnover based upon sediment profiles that require decades or centuries to develop.

We assumed that the amount of mineralized N lost from the cohort is proportional to the quantity of soluble N. Thus, the amount of labile organic N in the cohort changes as a result of additions due to root turnover as well as losses from the soluble pool:

$$N_L(t+1) = N_L(t) + k_2 k_7 (1 - f_n) W_b(t) - k_6 r G(t). \quad [17]$$

Similarly for refractory organic N

$$N_R(t+1) = N_R(t) + k_2 k_7 f_n W_b(t) \quad [18]$$

where k_2 and k_7 are the turnover rate (yr^{-1}) and N content (g N g dry wt^{-1}) of live belowground biomass, respectively, and r and k_6 are the residence time (yr) and specific loss rate (yr^{-1}) of soluble N.

The total quantity of particulate organic N (N_T) within the sediment cohort was defined as the sum of labile (N_L) and refractory (N_R) organic N pools and organic N in live roots

$$N_T(t) = N_L(t) + N_R(t) + k_7 W_b(t). \quad [19]$$

The total annual export of N (M_n) from the sediment (g

$\text{N cm}^{-2} \text{ yr}^{-1}$) between the surface (at $t = 0$) and a given depth corresponding to time $= n$ is calculated by summing the loss term in Eq. [17]

$$M_n = \sum_{t=0}^n k_6 r G(t) \quad [20]$$

while the total input of organic N to the sediment (I_n) is equivalent to the sum of surface and root inputs

$$I_n = N_T(0) + \sum_{t=0}^n k_2 k_7 W_b(t). \quad [21]$$

Note that $G(t)$ and $W_b(t)$ are the mineralization rate and root biomass within the cohort of age t and that these terms reduce to zero as time increases, i.e., as sediment depth increases.

The concentration of soluble N in the pore water within the cohort can be calculated as:

$$N(t) = rG(t) / [V(t) - C_T(t)/k_8 - W(t)/k_9] \quad [22]$$

where k_8 and k_9 are the particle densities (g dry wt cm^{-3}) (Blake, 1965) of organic and inorganic matter, respectively.

We assumed, for argument, that the rate of mineralization of other nutrients, like P (P_o), would be proportional to rates of mineralization and loss of N. That is:

$$P_o(t+1) = P_o(t) + k_{10} k_2 W_b(t) - k_6 r G(t) / k_{11} \quad [23]$$

where k_{10} is the concentration of P in belowground biomass (g P g dry wt^{-1}), and k_{11} is the ratio of N/P losses from the cohort. Initially, $P_o(0)$ is the annual deposition of organic P onto the marsh surface (g P cm^{-2}). We derived Eq. [23] as an exercise to determine if the sediment profile of organic P was related to that of N by some simple proportion, and to determine if this proportion (k_{11}) resembled the ratio of nutrients in the biota.

PARAMETER ESTIMATION

Of 22 constants in the model (Table 1), 13 appear in equations that generate sediment profiles of organic matter, six

Table 3. Combinations of independent (denoted by †) and dependent parameter values that gave acceptable agreement between model calculations and observed sediment nutrient profiles.‡

f_c †	k_2 †	R_f †	f_n	k_4	k_6	m	k_{11} §
0.2	0.5	0.1	0.45	0.02	12.	1.09	8.2
0.2	0.5	0.2	0.38	0.027	23.	0.968	8.55
0.2	1.0	0.1	0.38	0.027	23.	0.968	8.55
0.1	0.5	0.1	0.225	0.01	27.	0.516	9.55

‡ The top row of parameter estimates were accepted as the best.

§ Symbols as in Tables 1 and 2.

appear in equations that generate TON profiles, and the remaining three appear in equations that generate TOP profiles. Whenever possible, we used independent measurements as values for the parameters in the model. Sources for these independent parameter values, given in Table 1, included our own direct measurements plus reliable literature values, or in one case, an assumption, and are discussed in subsequent sections.

Values for some constants (termed dependent parameters) were unknown, but were estimated by fitting the model to the observed profiles of sediment organic matter, TON and TOP. For the simulations reported here, six parameters were unknown or dependent; these are identified in Table 1 as the parameters whose values gave a best fit of the model. For other applications the identity and number of dependent and independent parameters may change. For example, we assigned an independent value to the refractory fraction of organic matter input (f_c) and then estimated the allochthonous input of inorganic matter (k_4 , used here as a dependent parameter) by fitting the model to the sediment profile of organic matter. However, if k_4 is known, then f_c could be estimated as a dependent parameter.

In principle, the model can be fitted to sediment profiles of organic matter; N and P for the purpose of estimating any unknown parameter, provided there is a unique solution, i.e., one unknown parameter that uniquely determines the fit of the model to a particular data set. In practice, however, the quality of the independent parameter estimates varies. We did a sensitivity analysis of the model with those parameters that we believe are least well known for our study site. Independent parameter estimates that we varied in this sensitivity analysis were the root biomass (R_f), root turnover rate (k_2), and refractory fraction of organic matter (f_c). These parameters were varied 100% and dependent parameters were adjusted (Table 3) to achieve the best fit of the model for combinations of independent parameter estimates. Given these data limitations, it is possible to define a range of values for a given parameter.

The effects of changes in belowground biomass (R_f) and turnover (k_2) on profiles of sediment organic matter, TON and TOP could be minimized by adjusting the values of the dependent parameters within reasonable ranges. However, we found that changes in the value of f_c produced the greatest changes in the simulated profiles. Profiles generated when $f_c = 20\%$, our best estimate, are presented in subsequent figures as dotted lines, those generated when $f_c = 10\%$ appear as a dash-dot line.

RESULTS

Dynamics of Organic Matter

The constants k and f_c in the decomposition equations (Eq. [1] and [5]) were determined directly from a litter bag study of the three dominant macrophyte species (*Typha*, *Carex*, and *Calamagrostis*) in the North River Marsh. Decomposition was rapid and similar for the three species. The litter showed sea-

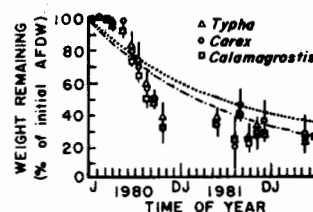


Fig. 2. Observed litter decomposition (± 1 SD) of the three dominant species at the study site and the simulated decomposition ($C_f(t)/C_f(0)$, exclusive of root inputs in Eq. [8]) for simulations in which $f_c = 0.2$ (.....) and in which $f_c = .1$ (-.-.-.-.-).

sonal variations in specific decay rate (Fig. 2), but these are unimportant for simulations of sediment profiles of organic matter because the time scales for decay of labile organic matter and sediment formation differ greatly. The litter lost 60 to 70% of its original dry weight after 1 yr, but lost little weight thereafter (Fig. 2). Our analysis indicated that 18.2 to 22.7% of the original weight of *Typha*, *Carex*, and *Calamagrostis* was resistant to decay, depending on species (Morris and Lajtha, 1986). Similar estimates for the fraction of primary production permanently incorporated into sediments have been made for other marshes (Hatton et al., 1982; Jordan et al., 1983; Smith et al., 1983). Figure 2 shows the simulated litter decomposition when the refractory fraction of the original litter (f_c) was assigned values of 20 and 10%.

Given values for the constants k and f_c (Table 1), the simulated profile of organic matter (Fig. 3) was dependent only on the constants that define the total inputs of organic and inorganic matter. The input of organic matter to the sediment surface was estimated independently to be $1 \text{ kg dry wt m}^{-2} \text{ yr}^{-1}$ from harvest measurements of standing biomass. Independent estimates of the amount of live belowground biomass (R_f) were based on data from cultures of *Typha*, *Carex*, and *Calamagrostis*. However, these cultures were not continued for > 1 yr and we have no measurement of the steady state turnover of roots and rhizomes. Considering that the ratio of above/belowground biomass in these cultures was about 1.5:1, and that rhizomes comprised about 50% of the belowground biomass (J. T. Morris, unpublished data) and probably turnover less frequently than roots, our best estimate of the turnover time of belowground biomass (k_2) is 0.5 yr^{-1} . We tested the sensitivity of the model to this value by varying root biomass and turnover 100% and

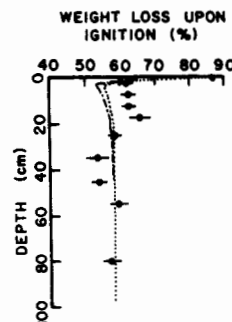


Fig. 3. Observed profile of sediment organic matter (weight loss upon ignition, ± 1 SE) and the simulated profile ($C_f(t)$ in Eq. [8]) vs. D_s in Eq. [12] when $f_c = 0.2$ (.....) and when $f_c = 0.1$ (-.-.-.-.-).

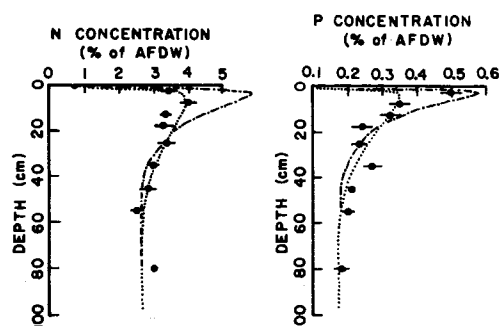


Fig. 4. Observed profiles of sediment organic N and P (± 1 SE) and the simulated profile (N_T/C_T in Eq. [19] and [8], and P_o/C_T in Eq. [23] and [8] vs. D_s in Eq. [12]) when $f_c = 0.2$ (.....) and when $f_c = 0.1$ (-.-.-.-).

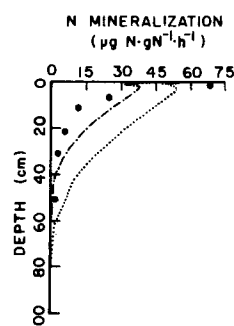


Fig. 5. Observed sediment profile of the rate of mineralization of organic N and the simulated profile (G/N_T in Eq. [16] and [19] vs. D_s in Eq. [12]) when $f_c = 0.2$ (.....) and when $f_c = 0.1$ (-.-.-.-).

then adjusting several dependent parameters (Table 3) to obtain an acceptable fit of the model.

Finally, the value of the input rate of exogenous inorganic matter was found by fitting the model to the observed sediment profile of AFDW (Fig. 3). We found that, depending on values assigned to several independent parameters (Table 3), an input rate ranging from 0.010 to 0.027 g cm⁻² yr⁻¹ of inorganic material (k_4) gave acceptable agreement with the observed data.

The simulated and observed AFDW decreased quickly within the top 5 cm of sediment (Fig. 3) because of the rapid decomposition of organic matter. The calculated AFDW of about 85% at the sediment surface, which approximates the composition of the initial mixture of undecomposed plant litter and allochthonous inorganic material, decreased to a minimum of about 55% within the top 5 cm, then increased slightly between 5 and 30 cm to an asymptotic value of 59% (Fig. 3). When model calculations were averaged over 5-cm sections to simulate a typical field sampling protocol, the simulated mean AFDW was 63% in the top 5-cm section compared to an observed mean of 62%.

Dynamics of Nitrogen and Phosphorus

The profile of TON in sediment is a function of the decomposition rate, the net loss rate of N from sediment, and the relative input rates of inorganic matter and of labile and refractory organic C and N. Decomposition of organic matter and inputs of organic and inorganic matter were determined as described above. Net N export is a function of the amount of labile ($1 - f_n$) organic N, specific mineralization rate (m) of labile organic N, and the residence time (r) and specific loss rate (k_6) of soluble N. Values for these parameters (Table 1) were determined by fitting the model to the sediment profiles of TON (Fig. 4), N mineralization rates (Fig. 5), and inorganic N (Bowden 1984b; J. T. Morris, unpublished data).

The simulated profile of sediment TON quickly increased from 0.77% of AFDW at the sediment surface to 3.9% at a depth of 7 cm (when $f_c = 20\%$) and then gradually declined to an asymptotic value of 2.6% of AFDW (Fig. 4). The surface value approximates the concentration of TON in litter; the increase in TON between the sediment surface and 7 cm is due to the initial, rapid decay of labile organic C and the relatively slower mineralization of TON. At depths below

the zone of rapid C mineralization, TON concentration gradually decreased as TON continued to slowly mineralize. The observed profile of TON, from 5-cm core sections, increased from 3.5% in the top 5 cm to 4.0% between 5 and 10 cm, and then decreased to values in the range of 2.5 to 3.0% of AFDW (Fig. 4). This decrease in TON concentration with depth is a characteristic feature of other wetland soils (Coultas, 1978).

Both the observed and simulated profiles of gross N mineralization decreased approximately exponentially with sediment depth from about 70 $\mu\text{g N g N}^{-1} \text{ h}^{-1}$ at the surface to near zero at the 60-cm depth (Fig. 5). Similar depth dependent trends in N mineralization have been observed in terrestrial soils (Cassman and Munns, 1980; Federer, 1983). However, the simulated mineralization rates generally exceeded the observed rates, which were based on an ¹⁵N isotope dilution technique (Bowden, 1984a). There are several possible reasons for this discrepancy including temporal and spatial variations in the measured mineralization rates and differences in the operational definitions of mineralization. The ¹⁵N technique measured gross ammonification rates in core sections incubated in the laboratory while the model calculates the mean annual, gross rate of solubilization of organic N.

The sediment profile of TOP was simulated by assigning an appropriate value to the ratio of N/P loss (k_{11}). Depending on values assigned to other parameters (Table 3), ratios of 8.20 to 9.55 (g g⁻¹) gave an acceptable fit to the observed TOP profile (Fig. 4). For $f_c = 20\%$, the simulated TOP profile increased from 0.07% of AFDW at the sediment surface, which approximates the concentration in litter, to 0.35% at the 5-cm depth. This increase resulted because TOP was concentrated, like TON, during the initial phase of rapid decomposition of labile organic C. Simulated TOP concentrations decreased below 7 cm and reached an asymptotic value of 0.17% of AFDW near the 60-cm depth (Fig. 4). Only slight changes were noted in both the simulated and observed TOP concentrations between 40 and 80 cm.

The N/P ratio of sediment organic matter increased with depth to a maximum value of about 15:1 (g/g) at the 50-cm depth (Fig. 6). Similar trends were observed by Hatton et al. (1982) in Louisiana marshes. The model correctly simulated the overall trend of an increase in the sediment N/P ratio to a level of about

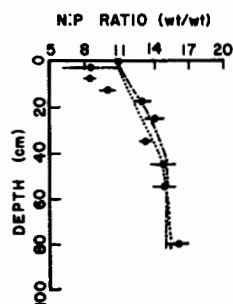


Fig. 6. Observed sediment profile of the weight ratio of organic N/organic P (± 1 SE) and the simulated profile (N_T/P_o in Eq. [19] and [23] vs. D_s in Eq. [12]) when $f_c = 0.2$ (.....) and when $f_c = 0.1$ (-----).

15:1. However, the model failed to simulate an initial decrease in the N/P ratio from 11:1, which approximates the composition of plant litter, to a low of about 8.5:1 in the 5- to 10-cm depth (Fig. 6). This initial decrease in the N/P ratio within surface sediments was due to an enrichment of TOP relative to TON. This was also seen in decomposing litter, as discussed below.

Increases in concentrations of TON and TOP in surface sediments (Fig. 4) to levels above those observed in the initial litter input are consistent with both predicted and observed increases of TON and TOP in the decomposing litter of the dominant macrophytes (Fig. 7). After 2.5 yr of decomposition, concentrations of TON in litter increased from an initial value of 0.7% to about 2.2% of AFDW (Fig. 7). The model correctly simulated this trend and also predicted that TON concentrations would have continued to increase to the maximum levels observed in surface sediments.

Concentrations of TOP in litter, while more variable than TON, increased from about 0.07% to levels between 0.25 and 0.7% of AFDW after 2.5 yr, depending on species, whereas the predicted concentration at 2.5 yr was about 0.2% (Fig. 7). The model underestimated the concentration of TOP in decomposing litter (Fig. 7) because of a large net immobilization of TOP by litter. Immobilization of TOP was relatively greater than TON, and decomposing litter maintained $> 100\%$ of its original TOP content for as much as 2 yr following the onset of decomposition (Morris and Lajtha, 1986). Consequently, the N/P ratio in litter decreased during the 2.5-yr litter bag study. This is consistent with the decrease in N/P ratio from 11:1 in the initial litter to 8.5:1 in the surface sediment (Fig. 6).

DISCUSSION

We developed a process-oriented model of sedimentation and organic matter diagenesis based largely on independent measurements made in the North River Marsh. The model serves three purposes. First, it provides a means to calculate various rate processes, such as annual nutrient turnover. We relied on information inherent in the depth distributions of sediment organic matter, TON, and TOP, to infer values for nutrient mineralization rates. Second, it provides a means to explore the consequences of altering var-

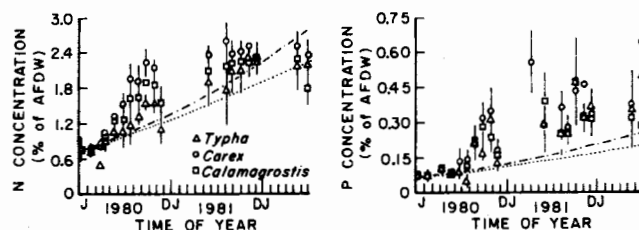


Fig. 7. Observed concentrations of organic N and P (± 1 SD) in decomposing litter of the three dominant species and the simulated concentrations (N_T/C_T or P_o/C_T in Eq. [19], [8], and [23], exclusive of root inputs) when $f_c = 0.2$ (.....) and when $f_c = 0.1$ (-----).

ious rates, which is useful for both management and basic scientific purposes. In some cases where independent parameter estimates are poorly defined we have presented several scenarios that define a range of expected results. Finally, the model provides a means to standardize comparisons between different sediment or soil systems.

The model clearly demonstrates that loss of organic C from litter is very rapid and exceeds initial losses of TON and TOP. The mineralization of labile organic C compounds from surface litter inputs was predicted to be essentially complete within the top 2 or 3 cm (Fig. 3). This conclusion is consistent with the measured depth profile of AFDW which did not show any clear trends below the surface (Fig. 3). The model predicted a slight increase in AFDW beginning near the sediment surface to a depth of about 25 cm due to the input of refractory organic matter associated with root turnover.

The profiles of TON and TOP in sediment can be explained by the relative differences in the rates of mineralization and export of these nutrients and C. Simulated profiles of TON and TOP increased initially at the sediment surface, from values characteristic of the litter, to maximum values just below the surface (Fig. 4) due to a rapid loss of labile organic C in this zone. Nitrogen mineralization within the top 60 cm of sediment (Fig. 5) resulted in a gradual decrease in TON concentration beneath the surface zone of rapid C loss (Fig. 4). The average mineralization rate of TON within the top 60 cm is consistent with rates reported for flooded soils elsewhere (Rao et al., 1984).

Sediments deeper than 60 cm apparently contained only refractory N at concentrations of about 2.5 to 3% of AFDW (Fig. 4). This is within the range of values reported for the composition of humic acids (Martin and Haider, 1971; Schnitzer, 1977; Cortez and Schnitzer, 1979). Model predictions of the refractory TON concentration at this depth depend on the relative rates of input of refractory organic C (f_c) and refractory TON (f_n). Litter bag studies provided an estimate of 20% for f_c (Morris and Lajtha, 1986). Model simulations with $f_c = 10$ and 20% required that $f_n = 22.5$ and 45%, respectively (Table 3), to produce TON distributions consistent with the observed profile (Fig. 4).

In litter, the increase in concentration of TON and TOP was initially greater than that which can be accounted for by losses of labile C. This net immobili-

zation of nutrients could be due to both biotic and abiotic factors [see Odum et al. (1979) and their citations]. Relatively more TOP was immobilized than TON; for example, *Carex* litter contained as much as 275% of its original TOP content after several months of decomposition compared to about 175% of its original TON (Morris and Lajtha, 1986). This relatively greater and longer immobilization of TOP may account for the difference between the N/P ratio in the original litter (10:1–13:1 depending on species) and the ratio that was observed in the top 10 cm of sediment (8.5:1).

The amount of N required by primary producers made available through mineralization of soil-TON depends on the relative fractions of labile and refractory (f_n) TON in the litter input and the age of the sediment. For a 1-m sediment column in the North River Marsh, the total annual N loss from sediments, from Eq. [20], was calculated to be 9.9 to 17.5 g N m⁻² yr⁻¹ or 55 to 78% of the annual TON input, depending on values assigned to f_n , k_2 , and R_l (Table 3). These are the maximum annual N loss rates for this marsh, given by $N(0) + k_2 k_7 (1 - f_n) R_l$, and are characteristic of a mature sediment in which sufficient time had elapsed for labile organic N to completely mineralize. A shallow sediment ($\ll 1$ m) would have a lower N loss, provided that its age did not exceed the time required for complete N mineralization. For example, sediments with maximum depths of only 10 to 15 cm would have loss rates 50% lower than those predicted for a 1-m sediment column.

The model predicted that a greater percentage of the input of TOP is turned over annually than is TON. Net TOP turnover was estimated to be in the range 1.19 to 2.02 g P m⁻² yr⁻¹ or 71 to 85% of the total input rate. Fractions of labile and refractory TOP were not explicitly defined in the model; instead, TOP mineralization was assumed to be directly proportional to TON mineralization.

Model simulations indicate that the ratio of N/P loss (k_{11}) is a constant that has a value of 8.20 to 9.55, depending on assumptions regarding other parameter values (Table 3). Redfield (1958) proposed that N and P cycle in the marine environment in a ratio of about 7 N/1 P (g/g), determined by the average composition of phytoplankton. Analogously, the ratios in which nutrients cycle through plant-soil systems are determined in large part by the chemical composition of the plants. The aboveground plant litter had an average N/P ratio of about 11:1 which is reasonably close to the values assigned to k_{11} .

This model shows that our knowledge of sediment organic matter and nutrient dynamics in marshes is weak in several specific areas. First, it is a persistent problem in ecological research that there are few accurate estimates of belowground biomass or production. These two parameters have a large influence on calculations of nutrient mineralization and organic matter accumulation rates. Furthermore, it is equally important to better define the amounts and dynamics of refractory material in organic matter, particularly in the subsurface environment. Finally, it would be helpful to know the importance of belowground nutrient translocation from senescing roots and rhizomes.

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